

FORM VI MATHEMATICS EXTENSION 2

Time allowed: 3 hours (plus 5 minutes reading time) **Exam date:** 6th August 2003

Instructions:

- All questions may be attempted.
- All questions are of equal value.
- Part marks are shown in boxes in the right margin.
- All necessary working must be shown.
- Marks may not be awarded for careless or badly arranged work.
- Approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Collection:

- The writing booklets will be collected in one bundle.
- Start each question in a new writing booklet.
- If you use a second booklet for a question, place it inside the first. Don't staple.
- Write your candidate number on each booklet.

Checklist:

- SGS Writing Booklets required — eight booklets per boy.
- Candidature: 54 boys.

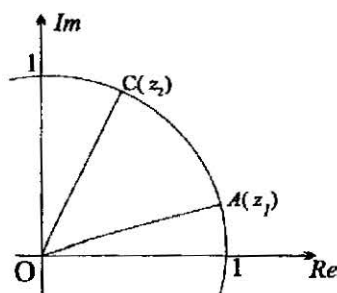
QUESTION ONE (Start a new writing booklet)

- | | Marks |
|---|--|
| (a) Find $\int \frac{\sin x}{\cos^5 x} dx$. | <div style="border: 1px solid black; padding: 2px; display: inline-block;">1</div> |
| (b) Use completion of squares to evaluate $\int_{-2}^{-1} \frac{5}{x^2 + 4x + 5} dx$. | <div style="border: 1px solid black; padding: 2px; display: inline-block;">3</div> |
| (c) (i) Find the real numbers A , B and C such that
$\frac{3x^2 - x + 8}{(1-x)(x^2 + 1)} \equiv \frac{A}{1-x} + \frac{Bx + C}{x^2 + 1}.$ | <div style="border: 1px solid black; padding: 2px; display: inline-block;">3</div> |
| (ii) Hence find $\int \frac{3x^2 - x + 8}{(1-x)(x^2 + 1)} dx$. | <div style="border: 1px solid black; padding: 2px; display: inline-block;">2</div> |
| (d) Use integration by parts to show that $\int_1^4 \frac{\ln x}{\sqrt{x}} dx = 4(2 \ln 2 - 1)$. | <div style="border: 1px solid black; padding: 2px; display: inline-block;">3</div> |
| (e) Use the substitution $t = \tan \frac{\theta}{2}$ to find $\int \frac{1}{1 + \cos \theta} d\theta$. | <div style="border: 1px solid black; padding: 2px; display: inline-block;">3</div> |

QUESTION TWO (Start a new writing booklet)

Marks

- (a) Find the square roots of $9 - 40i$. Give your answers in the form $a + ib$. 3
- (b) Sketch on the Argand diagram the locus $|z - 1| = |z + i|$. 1
- (c) Sketch the region in the Argand diagram that satisfies both the conditions 2
 $-\frac{\pi}{2} \leq \arg(z - 2) \leq 0$ and $\operatorname{Im}(z) \leq -1$.
- (d) Let $z = 1 - i$ and $w = -1 + i\sqrt{3}$. 1
- (i) Find $\arg z$ and $\arg w$. 1
- (ii) Hence find $\arg(wz)$. 1
- (iii) Hence prove that $\sin \frac{5\pi}{12} = \frac{\sqrt{2}(1 + \sqrt{3})}{4}$. 2
- (e) (i) Let $z = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$. Find z^9 . 1
- (ii) On the Argand diagram, plot and label all complex numbers that satisfy both 2
the conditions $z^9 = -1$ and $\operatorname{Re}(z) \leq 0$.
- (f)

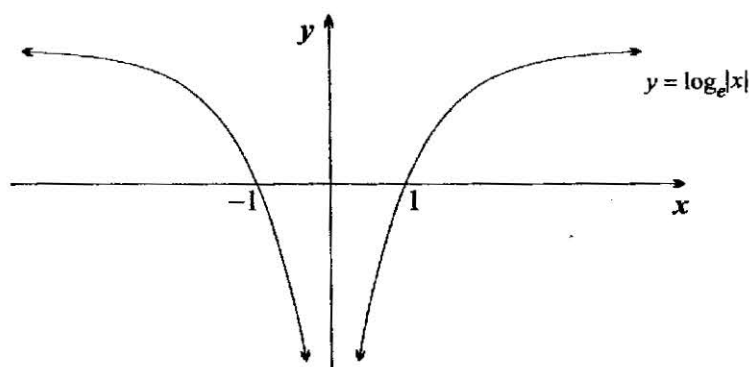


In the Argand diagram above, the two points A and C lie on the circumference of the circle with centre the origin and radius 1. They represent the complex numbers z_1 and z_2 respectively.

- (i) Copy the diagram into your answer booklet. Then mark on your diagram the position of the point B that represents the complex number $z_1 + z_2$. 1
- (ii) Explain why AC is perpendicular to OB . 1

QUESTION THREE (Start a new writing booklet)

(a)



The graph above shows the function $y = f(x) = \log_e |x|$.

Marks

(i) Use half a page to sketch on a number plane the graph $y = f\left(\frac{x}{2}\right)$.

1

(ii) Use half a page to sketch on a number plane the graph $y = \frac{1}{f(x)}$.

2

(iii) Use half a page to sketch on a number plane the graph $y^2 = f(x)$.

2

(iv) Use half a page to sketch on a number plane the graph $y = e^{f(x)}$.

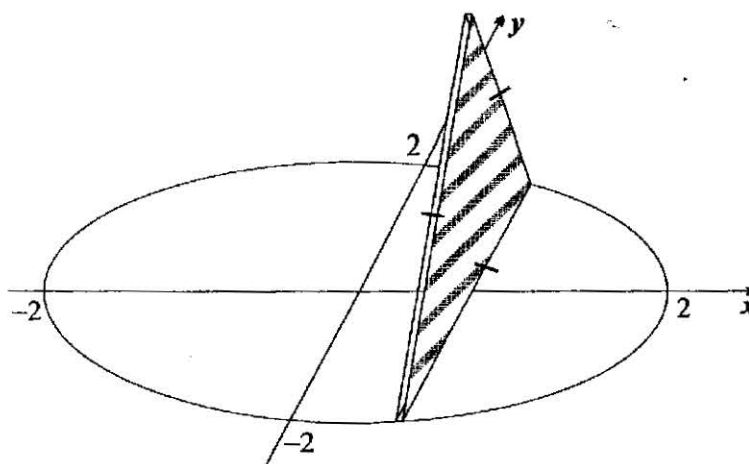
1

(b) Use the method of cylindrical shells to find the volume of the solid formed when the region bounded by the curve $y = x^2 + 3$ and the x -axis between the lines $x = 0$ and $x = 3$ is rotated about the y -axis.

3

(c)

3



The diagram above shows a cross-sectional slice of a solid whose base is the region enclosed by the circle $x^2 + y^2 = 4$. Each such cross-section of the solid is an equilateral triangle. Find the volume of the solid.

(d) The region between the curve $y = \sin x$ and the line $y = 1$, from $x = 0$ to $x = \frac{\pi}{2}$, is rotated around the line $y = 1$. Using a slicing technique find the volume formed.

3

QUESTION FOUR (Start a new writing booklet)

- (a) A mass of 2 kg, on the end of a string 0.6 metres long, is rotating as a conical pendulum, with angular velocity 3π radians per second. Take the acceleration due to gravity to be 10 m/s^2 .

Let θ be the angle that the string makes with the vertical.

Marks

- (i) Draw a diagram showing all forces acting on the mass.

1

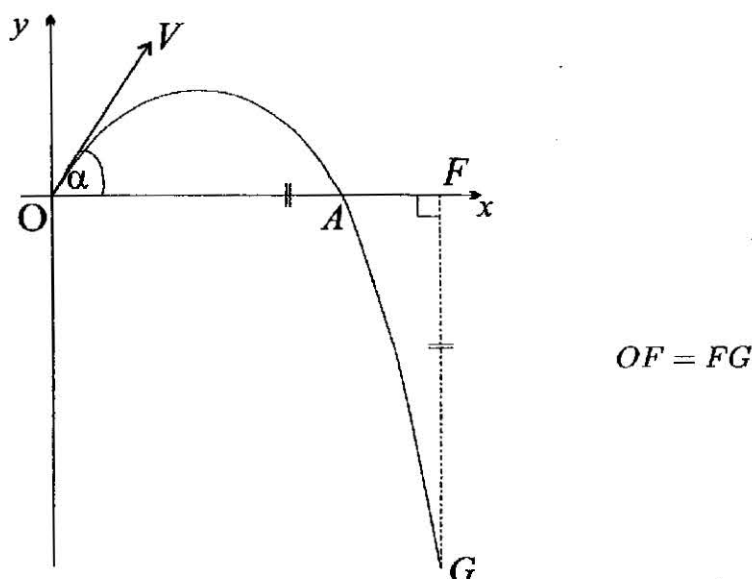
- (ii) By resolving forces, find the tension in the string.

3

- (iii) Find θ correct to the nearest degree.

1

(b)



In the diagram above, a projectile is fired from a point O at the top of a vertical cliff. Its initial speed is $V \text{ m/s}$ and its angle of elevation is α . Let the acceleration due to gravity be $g \text{ m/s}^2$.

- (i) By using the equations of motion $\ddot{x} = 0$ and $\ddot{y} = -g$, derive expressions for the horizontal and vertical displacements after t seconds.

2

- (ii) Let G be the point on the projectile's path where the distance below the origin equals the distance to the right of the origin. That is, $OF = FG$ on the diagram above.

- (a) Prove that the time taken for the projectile to reach G is

2

$$\frac{2V(\sin \alpha + \cos \alpha)}{g} \text{ seconds.}$$

(β) Show that $OF = \frac{V^2}{g}(\sin 2\alpha + \cos 2\alpha + 1)$ metres. 2

(γ) Let A be the point on the projectile's path where it is level with the point of projection. If $OF = \frac{4}{3}OA$, find α , correct to the nearest degree. 4

You may assume that the distance OA is given by $OA = \frac{V^2 \sin 2\alpha}{g}$ metres.

QUESTION FIVE (Start a new writing booklet)

Marks

(a) (i) Find the general solution of $\tan 4\alpha = 1$. 1

(ii) Use the binomial theorem and de Moivre's theorem to show that 4

$$\tan 4\alpha = \frac{4 \tan \alpha - 4 \tan^3 \alpha}{1 - 6 \tan^2 \alpha + \tan^4 \alpha}.$$

(iii) Hence solve the equation $x^4 + 4x^3 - 6x^2 - 4x + 1 = 0$. 3

(iv) Hence show that 3

$$\tan^2 \frac{\pi}{16} + \tan^2 \frac{3\pi}{16} + \tan^2 \frac{5\pi}{16} + \tan^2 \frac{7\pi}{16} = 28.$$

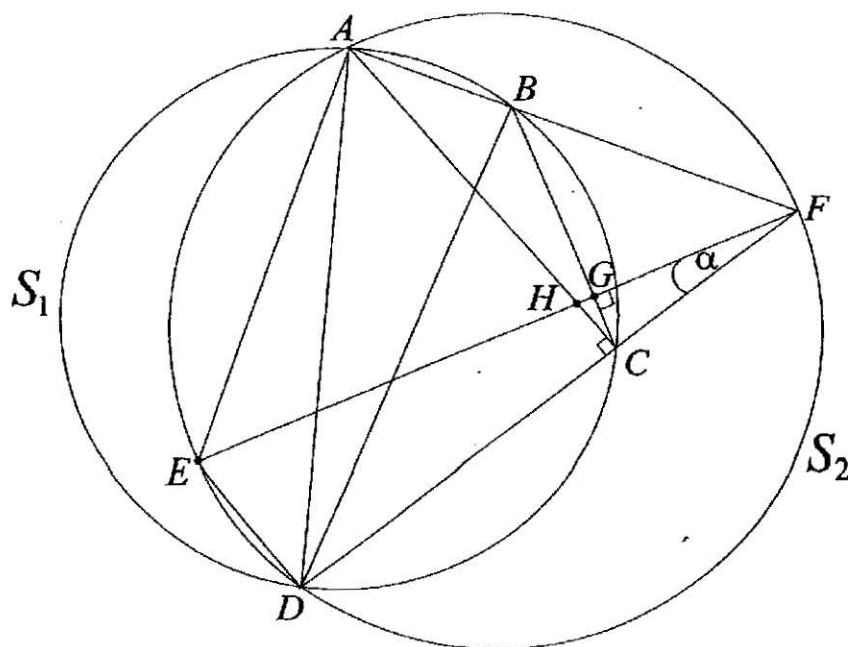
(b) Let α , β and γ be the roots of the equation $x^3 + px^2 + qx + r = 0$.

(i) Show that if the roots form an arithmetic sequence, then $2p^3 - 9pq + 27r = 0$. 2
HINT: If α , β and γ form an arithmetic sequence, then $\alpha + \beta + \gamma = 3\beta$.

(ii) Find a similar identity involving p , q and r that holds if the roots form a geometric sequence. 2

QUESTION SIX (Start a new writing booklet)

(a)



In the diagram above, $ABCD$ is a cyclic quadrilateral inscribed in the circle S_1 , and $AC \perp DC$.

The chords AB and DC produced intersect at F , and S_2 is the circle through A , D and F .

The line through F perpendicular to BC meets BC at G , meets AC at H and meets the circle S_2 at E .

Let $\angle DFE = \alpha$.

Marks

1

1

2

1

(i) Prove that $\angle HCG = \alpha$.

(ii) Prove that $AB \perp DB$.

(iii) Prove that $AE \parallel BD$.

(iv) Prove that E , A , B and G are concyclic.

(b) Let ω be one of the non-real cube roots of 1.

1

(i) Show that $1 + \omega + \omega^2 = 0$.

2

(ii) Hence find the value of $(2 - \omega)(2 - \omega^2)(2 - \omega^4)(2 - \omega^5)$.

(c) An object of mass 20 kg is dropped in a medium where the resistance at speed v m/s has a magnitude of $2v$ newtons. The acceleration due to gravity is 10 m/s^2 .

1

(i) Draw a diagram to show the forces on the object and show that the equation of

motion is $\ddot{x} = \frac{100 - v}{10}$.

- (ii) Find an expression for the velocity at time t seconds after the object is dropped. [2]
 (iii) Find the terminal velocity of the object. [1]
 (iv) Show that the distance x metres travelled when the speed is v m/s is given by [2]

$$x = 1000 \log_e \left(\frac{100}{100 - v} \right) - 10v.$$

- (v) Hence find the distance the object has fallen before reaching half its terminal velocity. [1]

QUESTION SEVEN (Start a new writing booklet)

- (a) A straight line is drawn through a fixed point $P(a, b)$ in the first quadrant on a number plane. The line cuts the positive part of the x -axis at A and the positive part of y -axis at B . Let $\angle OAB = \theta$.

Marks

- (i) Prove that the length of AB is given by [2]

$$AB = a \sec \theta + b \operatorname{cosec} \theta.$$

- (ii) Show that the length of AB will be a minimum if [3]

$$\cot \theta = \left(\frac{a}{b} \right)^{\frac{1}{3}}.$$

- (iii) Show that the minimum length of AB is $\left(a^{\frac{2}{3}} + b^{\frac{2}{3}} \right)^{\frac{3}{2}}$. [2]

- (b) (i) On the same number plane, sketch the graphs $y = \pi \sin x$ and $y = x$, for $0 \leq x \leq \pi$. [1]

- (ii) Explain why there is a number α between 0 and π such that $\pi \sin \alpha = \alpha$. Furthermore, show that $\frac{2\pi}{3} < \alpha < \frac{3\pi}{4}$. Do NOT try to evaluate α . [1]

- (iii) Let $f(x) = \sqrt{\pi^2 - x^2} \cos x - x \sin x$, for $-\pi \leq x \leq \pi$.

- (α) Prove that $f(x)$ is an even function. [1]

- (β) Evaluate $f(x)$ at $x = 0$, $\frac{\pi}{3}$, $\frac{\pi}{2}$ and π . [1]

- (γ) If α is the number defined in part (ii), show that $f(\alpha) = -\pi$. [1]

- (δ) Show that $f'(\alpha) = 0$, and hence find 3 stationary points of $f(x)$ and determine their nature. [3]

QUESTION EIGHT (Start a new writing booklet)

Marks

- (a) (i) Find
- k
- in terms of
- n
- if
- $\sin n\theta + \sin(n-2)\theta = 2 \sin k\theta \cos \theta$
- .
- 1**

- (ii) If
- n
- is an integer greater than 1 and
- $I_n = \int \sin n\theta \sec \theta d\theta$
- , prove that
- 2**

$$I_n + I_{n-2} = \frac{2 \cos(n-1)\theta}{1-n} + C, \text{ where } C \text{ is a constant of integration.}$$

- (iii) Hence prove that
- $\int_0^{\frac{\pi}{2}} \frac{\cos 5\theta \sin \theta}{\cos \theta} d\theta = \frac{23}{15}$
- .
- 4**

- (b) (i) Let
- $a_1, a_2, \dots, a_{k+1}$
- be positive real numbers. Define the function
- $\psi(x)$
- by
- 3**

$$\psi(x) = a_1 + a_2 + \dots + a_k + x - (k+1)(a_1 a_2 \dots a_k x)^{\frac{1}{k+1}}, \text{ for } x > 0.$$

Show that the minimum value of $\psi(x)$ occurs at $x = x_0$, where

$$x_0 = (a_1 a_2 \dots a_k)^{\frac{1}{k}}.$$

- (ii) Let
- $A_n = \frac{a_1 + a_2 + \dots + a_n}{n}$
- and
- $G_n = \sqrt[n]{a_1 a_2 \dots a_n}$
- . By considering
- $\psi(a_{k+1})$
- 5**
-
- from part (i) and using mathematical induction, prove that
- $A_n \geq G_n$
- .

REP

$$\begin{aligned}
 1. \quad (a) \quad & \int \frac{\sin x}{\cos^5 x} dx \\
 &= \int \sin x \cos^{-5} x dx \\
 &= \frac{1}{4} \cos^{-4} x + c \\
 &= \frac{1}{4} \sec^4 x + c \\
 &\equiv \frac{1}{4 \cos^4 x} + c \quad \boxed{1}
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad & \int_{-2}^{-1} \frac{5}{x^2 + 4x + 5} dx \\
 &= \int_{-2}^{-1} \frac{5}{(x+2)^2 + 1} dx \\
 &= \left[5 \tan^{-1}(x+2) \right]_{-2}^{-1} \\
 &= 5(\tan^{-1} 1 - \tan^{-1} 0) \\
 &= \frac{5}{4} \pi \quad \boxed{3}
 \end{aligned}$$

$$(c) \quad (i) \quad \frac{3x^2 - x + 8}{(1-x)(x^2+1)} = \frac{A}{1-x} + \frac{Bx+C}{x^2+1}$$

$$\text{Hence } A = \frac{3-1+8}{1^2+1} = 5$$

$$3x^2 - x + 8 \equiv 5(x^2 + 1) + (Bx + C)(1 - x)$$

$$\text{Hence } 5 - B = 3 \text{ and } 5 + C = 8$$

$$\text{So } B = 2 \text{ and } C = 3 \quad \boxed{3}$$

$$\begin{aligned}
 (ii) \quad & \int \frac{3x^2 - x + 8}{(1-x)(x^2+1)} dx = \int \frac{5}{1-x} + \frac{2x+3}{x^2+1} dx \\
 &= \int \frac{5}{1-x} + \frac{2x}{x^2+1} + \frac{3}{x^2+1} dx \\
 &= \ln|x^2+1| - 5 \ln|1-x| + 3 \tan^{-1} x + c \\
 &= \ln \left| \frac{x^2+1}{(1-x)^5} \right| + 3 \tan^{-1} x + c \quad \boxed{2}
 \end{aligned}$$

$$\begin{aligned}
 (d) \quad & \int_1^4 \frac{\ln x}{\sqrt{x}} dx \\
 &= \left[2\sqrt{x} \ln x \right]_1^4 - 2 \int_1^4 \frac{\sqrt{x}}{x} dx \\
 &= 4 \ln 4 - 2 \int_1^4 \frac{1}{\sqrt{x}} dx \\
 &= 4 \ln 4 - 4 \left[\sqrt{x} \right]_1^4 \\
 &= 4(2 \ln 2 - 1), \text{ as required.} \quad \boxed{3}
 \end{aligned}$$

$$\begin{aligned}
 \text{(e)} \quad & \int \frac{1}{1 + \cos \theta} d\theta \\
 &= \int \frac{2}{(1 + t^2) \left(1 + \frac{1-t^2}{1+t^2}\right)} dt \\
 &= \int \frac{2}{1 + t^2 + 1 - t^2} dt \\
 &= \int dt \\
 &= t + c \\
 &= \tan \frac{\theta}{2} + c \quad \boxed{3}
 \end{aligned}$$

$$\text{Let } t = \tan \frac{\theta}{2}$$

$$\text{Hence } \cos \theta = \frac{1 - t^2}{1 + t^2}$$

$$\text{Also } d\theta = \frac{2 dt}{1 + t^2}$$

2. (a) Let $z = x + iy$, hence $z^2 = 9 - 40i = (x + iy)^2$

$$\text{So } x^2 - y^2 + 2ixy = 9 - 40i$$

Equate real and imaginary parts.

$$\text{So } x^2 - y^2 = 9 \text{ and } xy = -20$$

$$\text{Hence } x^2 - \frac{400}{x^2} = 9$$

$$\text{So } x^4 - 9x^2 - 400 = 0$$

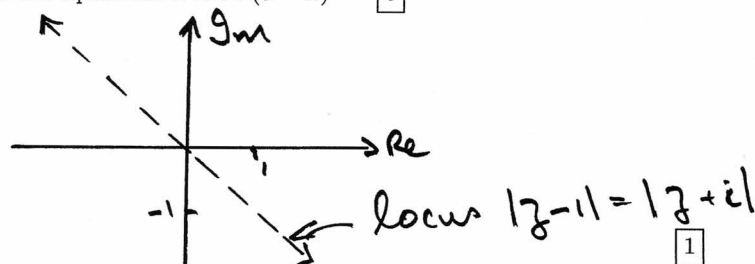
$$(x^2 - 25)(x^2 + 16) = 0$$

$$\text{But } x \in \mathbf{R}, \text{ so } x = \pm 5$$

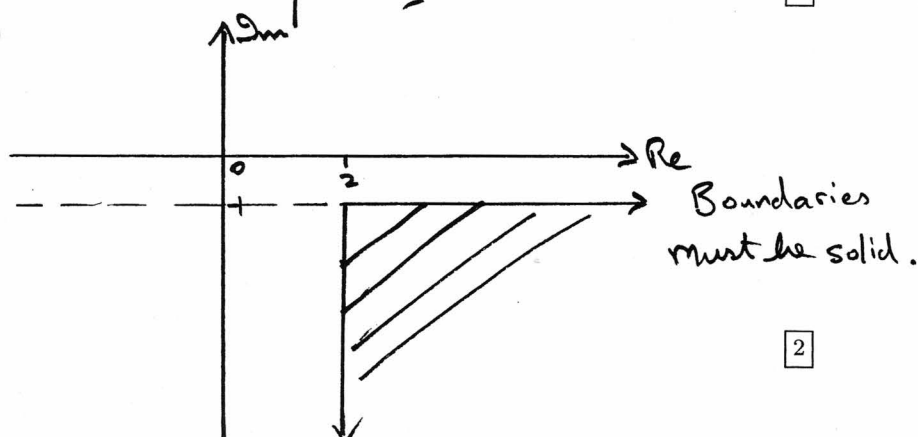
$$x = \pm 5 \text{ yields } y = \mp 4$$

$$\text{Hence the square roots are } \pm(5 - 4i) \quad \boxed{3}$$

(b)



(c)



(d) (i) $\arg z = -\frac{\pi}{4}$ and $\arg w = \frac{2\pi}{3}$ 1

(ii) $\arg(wz) = \arg w + \arg z = \frac{5\pi}{12}$ 1

(iii) Now $wz = \sqrt{3} - 1 + i(\sqrt{3} + 1)$

$$\text{Hence } \sin \frac{5\pi}{12} = \frac{\text{Im}(wz)}{|wz|} = \frac{\text{Im}(wz)}{|w||z|}$$

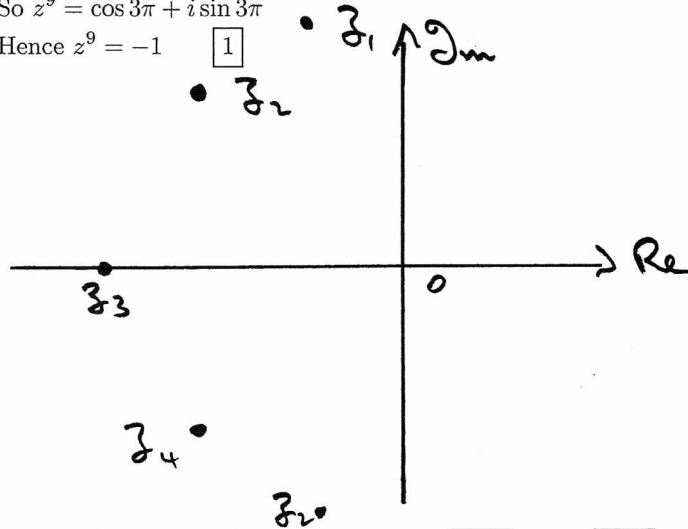
$$\text{So } \sin \frac{5\pi}{12} = \frac{\sqrt{3} + 1}{2\sqrt{2}} = \frac{\sqrt{2}(1 + \sqrt{3})}{4} \text{ as required.} \quad \text{2}$$

(e) (i) $z^9 = (\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})^9$

So $z^9 = \cos 3\pi + i \sin 3\pi$

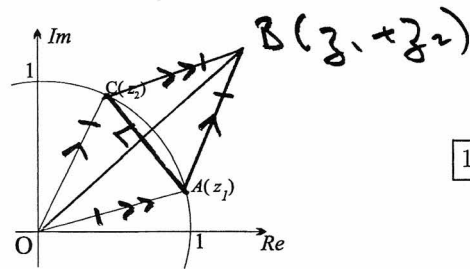
Hence $z^9 = -1$ 1

(ii)



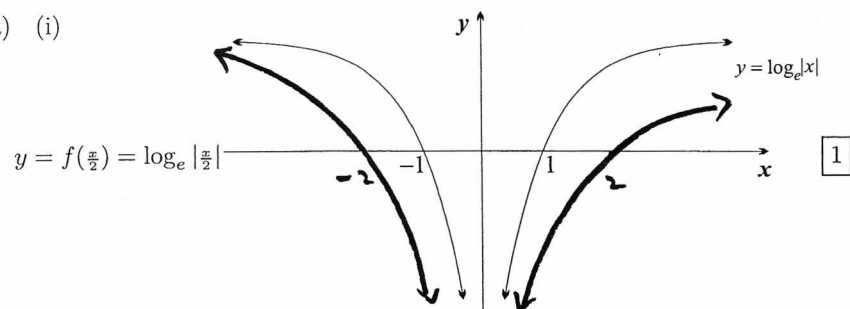
$$z_1 = \text{cis } \frac{5\pi}{9}, z_2 = \text{cis } \frac{7\pi}{9}, z_3 = -1, z_4 = \overline{\text{cis } \frac{7\pi}{9}}, z_5 = \overline{\text{cis } \frac{5\pi}{9}}. \quad \text{2}$$

(f) (i)

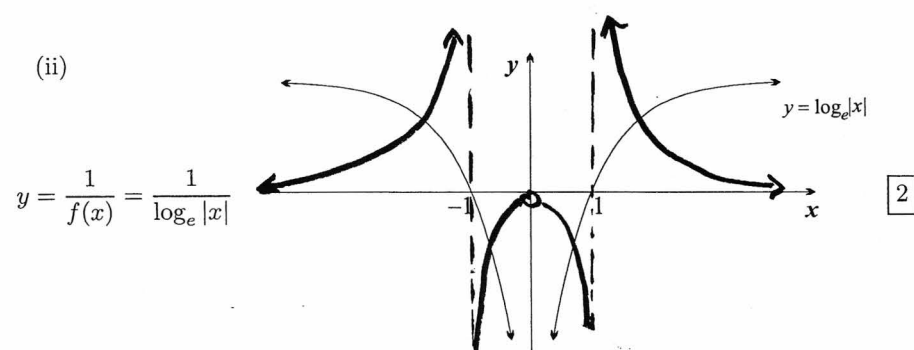


(ii) $OABC$ is a rhombus and hence the diagonals are perpendicular. 1

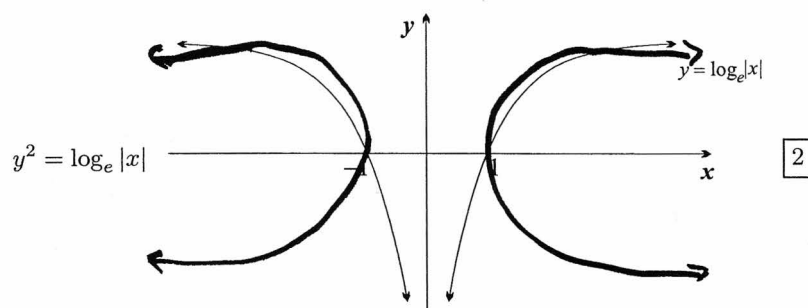
3. (a) (i)



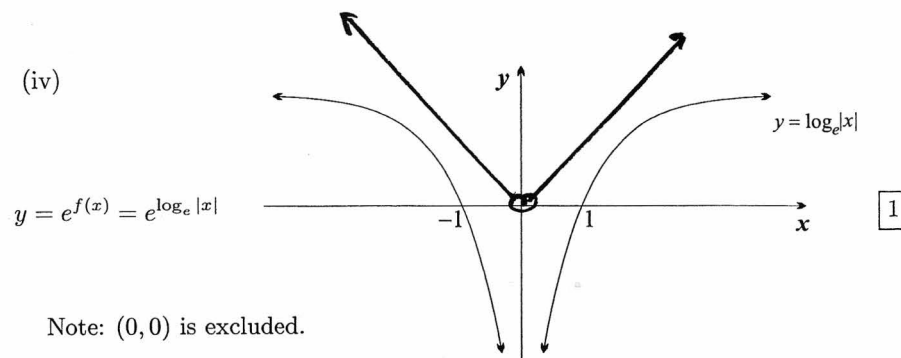
(ii)



(iii)

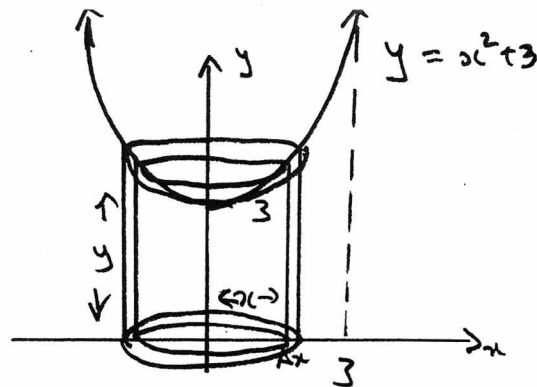


(iv)



Note: (0,0) is excluded.

(b)



The curved surface of each cylindrical shell is given by $SA = 2\pi xy = 2\pi(x^2 + 3)$.

Hence the volume of a shell Δx thick is $\approx 2\pi x(x^2 + 3)\Delta x$.

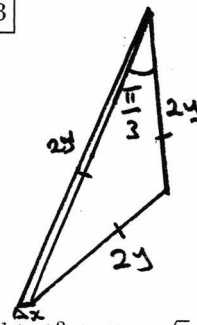
So the volume required is $V = 2\pi \int_0^3 x^3 + 3x \, dx$.

$$\text{So } V = 2\pi \left[\frac{1}{4}x^4 + \frac{3}{2}x^2 \right]_0^3$$

$$V = \frac{135}{2}\pi \text{ or } 67.5\pi \text{ units}^3.$$

3

(c)



Area of each cross-sectional slice is $\frac{1}{2}(2y)^2 \sin \frac{\pi}{3} = \sqrt{3}y^2$

Hence the volume of a slice Δx thick is $\approx \sqrt{3}y^2 \Delta x = \sqrt{3}(4 - x^2)\Delta x$.

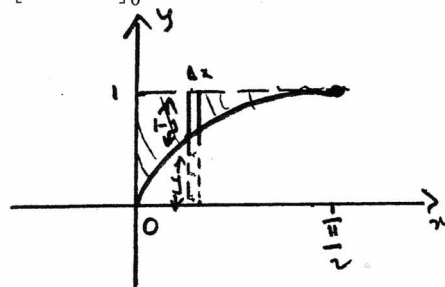
So the volume required is $\sqrt{3} \int_{-2}^2 4 - x^2 \, dx$.

$$\text{So } V = 2\sqrt{3} \int_0^2 4 - x^2 \, dx = 2\sqrt{3} \left(8 - \frac{8}{3} \right)$$

$$\text{So } V = \frac{32}{3}\sqrt{3} \text{ units}^3$$

3

(d)



The area of each slice of the solid is $\pi(1 - y)^2 = \pi(1 - \sin x)^2$.

If the slice is Δx thick then the volume is $\approx \pi(1 - \sin x)^2 \Delta x$.

$$V = \pi \int_0^{\frac{\pi}{2}} 1 - 2 \sin x + \sin^2 x \, dx$$

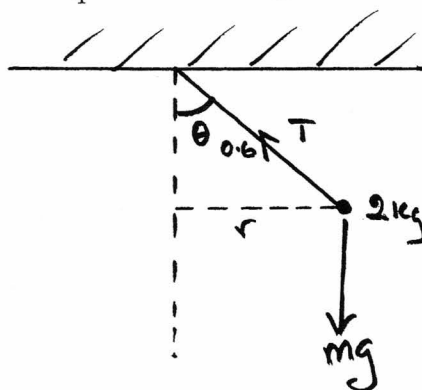
$$V = \int_0^{\frac{\pi}{2}} \frac{3}{2} - 2 \sin x - \frac{1}{2} \cos 2x \, dx$$

$$V = \left[\frac{3}{2}x + 2 \cos x - \frac{1}{4} \sin 2x \right]_0^{\frac{\pi}{2}}$$

$$V = \pi \left(\frac{3\pi}{4} - 2 \right)$$

$$\text{So } V = \frac{(3\pi - 8)\pi}{4} \text{ units}^3. \quad \boxed{3}$$

4. (a) (i)



$\boxed{1}$

(ii) Resolve forces at the mass.

$$\begin{array}{c} \text{vert} \\ \downarrow \end{array} T \cos \theta = 2g = 20$$

$$\begin{array}{c} \text{horoz} \\ \leftrightarrow \end{array} T \sin \theta = 2r\omega^2$$

$$\text{Hence } T \frac{r}{0.6} = 2r(3\pi)^2$$

$$\text{So } T = 2 \times 0.6 \times 9\pi^2$$

$$\text{i.e. } T = 10.8\pi^2 \approx 106.6 \text{ N} \quad \boxed{3}$$

$$\text{(iii) } \cos \theta = \frac{20}{T}$$

$$\text{So } \cos \theta = \frac{20}{10.8\pi^2}$$

$$\text{So } \theta = 79^\circ, \text{ to nearest } ^\circ. \quad \boxed{1}$$

(b) (i) $\ddot{x}(t) = 0$

Hence $\dot{x} = C_1$, a constant.

But $\dot{x}(0) = V \cos \alpha = C_1$.

Hence $\dot{x}(t) = V \cos \alpha$.

So $x(t) = V \cos \alpha t + C_2$,

where C_2 is a constant.

But $x(0) = 0 = C_2$.

Hence $x(t) = V \cos \alpha t$.

Also $\ddot{y} = -g$.

So $\dot{y} = -gt + C_3$,

where C_3 is a constant.

But $\dot{y}(0) = V \sin \alpha$,

Hence $C_3 = V \sin \alpha$.

So $\dot{y} = V \sin \alpha - gt$.

So $y = V \sin \alpha t - \frac{1}{2}gt^2 + C_4$,

where C_4 is a constant.

$y(0) = 0 = C_4$.

So $y(t) = V \sin \alpha t - \frac{1}{2}gt^2$.

↖ 2 ↗

(ii) (α) $OF = FG$ hence

$$V \sin \alpha t - \frac{1}{2}gt^2 = -V \cos \alpha t$$

$$\text{So } \frac{1}{2}gt = V \sin \alpha + V \cos \alpha, \quad (t \neq 0)$$

$$\text{So } t = \frac{2V(\sin \alpha + \cos \alpha)}{g} \text{ seconds.} \quad \boxed{2}$$

(β) $OF = V \cos \alpha t$

$$\text{So } OF = V \cos \alpha \frac{2V}{g} (\sin \alpha + \cos \alpha)$$

$$\text{So } OF = \frac{V^2}{g} (2 \sin \alpha \cos \alpha + 2 \cos^2 \alpha)$$

$$\text{So } OF = \frac{V^2}{g} (\sin 2\alpha + \cos 2\alpha + 1) \text{ m.} \quad \boxed{2}$$

(NOTE: Numerous solutions possible. The most common are below.)

(γ) $OF = \frac{4}{3}OA$, so $\frac{V^2}{g} (\sin 2\alpha + \cos 2\alpha + 1) = \frac{4}{3} \frac{V^2}{g} \sin 2\alpha$

$$\text{So } 3 \sin 2\alpha + 3 \cos 2\alpha + 3 = 4 \sin 2\alpha$$

$$\text{So } \sin 2\alpha - 3 \cos 2\alpha = 3.$$

$$\frac{1}{\sqrt{10}} \sin 2\alpha - \frac{3}{\sqrt{10}} \cos 2\alpha = \frac{3}{\sqrt{10}}$$

$$\text{Hence } \sin(2\alpha - \theta) = \frac{3}{\sqrt{10}},$$

$$\text{where } \cos \theta = \frac{1}{\sqrt{10}}$$

$$\text{and } \sin \theta = \frac{3}{\sqrt{10}}.$$

If $0^\circ \leq \theta \leq 90^\circ$ then $\theta = 71^\circ 34'$ to the nearest minute.

$$\text{So } 2\alpha = \sin^{-1} \left(\frac{3}{\sqrt{10}} \right) + \theta = 2\theta$$

That is $\alpha = \theta$.

OR Let $t = \tan \alpha$

$$\text{Hence } \sin 2\alpha = \frac{2t}{1+t^2}$$

$$\text{and } \cos 2\alpha = \frac{1-t^2}{1+t^2}$$

$$\text{So } \frac{2t}{1+t^2} - \frac{3(1-t^2)}{1+t^2} = 3$$

$$\text{So } 2t - 3 + 3t^2 = 3 + 3t^2$$

$$\text{So } 2t = 6$$

$$\text{So } \tan \alpha = 3$$

$$\text{So } \alpha = 72^\circ \text{ to the nearest degree.} \quad \boxed{4}$$

5. (a) (i) $\tan 4\alpha = 1$

So $4\alpha = n\pi + \frac{\pi}{4}, n \in \mathbb{Z}$

So $\alpha = (4n+1)\frac{\pi}{16}, n \in \mathbb{Z}$ 1

(ii) $(\cos \alpha + i \sin \alpha)^4 = \cos 4\alpha + i \sin 4\alpha$ (de M. th^m).

But the binomial theorem gives

$$(\cos \alpha + i \sin \alpha)^4 = \cos^4 \alpha + 4i \cos^3 \alpha \sin \alpha - 6 \cos^2 \alpha \sin^2 \alpha - 4i \cos \alpha \sin^3 \alpha + \sin^4 \alpha$$

Now equate the real and imaginary parts.

Hence $\sin 4\alpha = 4 \cos^3 \alpha \sin \alpha - 4 \cos \alpha \sin^3 \alpha$

and $\cos 4\alpha = \cos^4 \alpha - 6 \cos^2 \alpha \sin^2 \alpha + \sin^4 \alpha$

So $\tan \alpha = \frac{4 \cos^3 \alpha \sin \alpha - 4 \cos \alpha \sin^3 \alpha}{\cos^4 \alpha - 6 \cos^2 \alpha \sin^2 \alpha + \sin^4 \alpha}$

Hence $\tan 4\alpha = \frac{4 \tan \alpha - 4 \tan^3 \alpha}{1 - 6 \tan^2 \alpha + \tan^4 \alpha}$, as required. 4

(iii) $x^4 + 4x^3 - 6x^2 - 4x + 1 = 0$

So $4x - 4x^2 = x^4 - 6x^2 + 1$

i.e. $\frac{4x - 4x^3}{x^4 - 6x^2 + 1} = 1$

Let $x = \tan \alpha$

So $\frac{4 \tan \alpha - 4 \tan^3 \alpha}{1 - 6 \tan^2 \alpha + \tan^4 \alpha} = 1$

Hence $\tan 4\alpha = 1$

So $\alpha = (4n+1)\frac{\pi}{16}, n \in \mathbb{Z}$

Consider the values when $n = 0, \pm 1$ and -2 .

i.e. $x = \tan \frac{\pi}{16}, \tan \frac{5\pi}{16}, -\tan \frac{3\pi}{16}$ (or $\tan \frac{13\pi}{16}$) or $-\tan \frac{7\pi}{16}$ (or $\tan \frac{9\pi}{16}$) 4

(iv) $\tan^2 \frac{\pi}{16} + \tan^2 \frac{3\pi}{16} + \tan^2 \frac{5\pi}{16} + \tan^2 \frac{7\pi}{16}$

$= \left(\sum \alpha \right)^2 - 2 \sum \alpha \beta$

$= (-4)^2 - 2(-6)$

$= 28$, as required. 2

(b) (i) $\alpha + \beta + \gamma = 3\beta$

So $\beta = -\frac{p}{3}$

$\left(-\frac{p}{3}\right)^3 + p\left(-\frac{p}{3}\right)^2 + q\left(-\frac{p}{3}\right) + r = 0$

So $-p^3 + 3p^2 - 9pq + 27r = 0$

i.e. $2p^3 - 9pq + 27r = 0$ 2

(ii) $\alpha\beta\gamma = \beta^3$

So $\beta = \sqrt[3]{-r}$

Hence $\left(\sqrt[3]{-r}\right)^3 + p\left(\sqrt[3]{-r}\right)^2$

$+ q\left(\sqrt[3]{-r}\right) + r = 0$

So $-r + pr^{\frac{2}{3}} + q(-r)^{\frac{1}{3}} + r = 0$

i.e. $pr^{\frac{2}{3}} = qr^{\frac{1}{3}}$

So $p^3 r^2 = q^3 r$

Hence $p^3 r = q^3$ 2

6. (a) (i) $\angle GCD = \frac{\pi}{2} + \angle HCG = \frac{\pi}{2} + \alpha$ (Ext. $\angle \triangle CGF =$ sum of the int. opp. $\angle$'s)

Hence $\angle HCG = \alpha$, as required. 1

- (ii) $\angle ABD = \angle ACD = \frac{\pi}{2}$ ($\angle$'s in the same segment)

Hence $AB \perp DB$, as required. 1

- (iii) $\angle EAD = \alpha$ ($\angle$'s in the same segment)

$\angle ADB = \alpha$ ($\angle$'s in the same segment)

So $\angle BAD = \frac{\pi}{2} - \alpha$ ($\angle$ sum $\triangle BAD = \pi$)

Hence $\angle BAE = \alpha + \frac{\pi}{2} - \alpha = \frac{\pi}{2}$.

Hence $AB \perp AE$

So $AE \parallel BD$ (cointerior $\angle$'s are supplementary). 2

- (iv) $\angle BAE + \angle BGE = \frac{\pi}{2} + \frac{\pi}{2} = \pi$ ((iii) and given $FH \perp BC$)

Hence E, A, B and G are concyclic as the opposite $\angle$'s

are supplementary. 1

- (b) (i) $1 + \omega + \omega^2$ is a geometric series with common ratio ω .

$$\text{So } 1 + \omega + \omega^2 = \frac{\omega^3 - 1}{\omega - 1}$$

But $\omega^3 = 1$

Hence $1 + \omega + \omega^2 = 0$, as required. 1

- (ii) $(2 - \omega)(2 - \omega^2)(2 - \omega^4)(2 - \omega^5)$

$$= (2 - \omega)(2 - \omega^2)(2 - \omega)(2 - \omega^2), \text{ as } \omega^3 = 1$$

$$= ((2 - \omega)(2 - \omega^2))^2$$

$$= (4 - 2\omega - 2\omega^2 + \omega^3)^2$$

$$= (5 - 2(\omega + \omega^2))^2$$

But $\omega + \omega^2 = -1$ from (i).

$$\text{Hence } (2 - \omega)(2 - \omega^2)(2 - \omega^4)(2 - \omega^5) = (5 + 2)^2$$

$$\text{i.e. } (2 - \omega)(2 - \omega^2)(2 - \omega^4)(2 - \omega^5) = 49 \quad \text{[2]}$$

- (c) (i)

$\uparrow 2v$
 0
 $\downarrow$
 $20g$

Newton's 2nd law gives:

$$20\ddot{x} = 20g - 2v$$

$$\ddot{x} = 10 - \frac{v}{10}$$

$$\ddot{x} = \frac{100 - v}{10} \quad \text{[1]}$$

$$(ii) \quad \ddot{x} = \frac{dv}{dt} = \frac{100 - v}{10}$$

$$\text{So } \int \frac{dv}{100 - v} = \frac{1}{10} \int dt$$

$$\text{So } -\ln|100 - v| = \frac{t}{10} + c, \text{ for some constant } c.$$

$$\text{When } t = 0, v = 0$$

$$\text{hence } c = -\ln 100.$$

$$\text{So } -\frac{t}{10} = \ln \left| \frac{100 - v}{100} \right|$$

$$\text{So } 100e^{-\frac{t}{10}} = 100 - v$$

$$v = 100 \left(1 - e^{-\frac{t}{10}} \right) \quad \boxed{2}$$

(iii) Terminal velocity attained when either $t \rightarrow \infty$ or $\ddot{x} = 0$

$$\text{Hence the terminal velocity is } 100 \text{ m/s} \quad \boxed{1}$$

$$(iv) \quad \text{Now } \ddot{x} = \frac{100 - v}{10}$$

$$\text{So } v \frac{dv}{dx} = \frac{100 - v}{10}$$

$$\text{So } \frac{dv}{dx} = \frac{100 - v}{10v}$$

$$\text{So } \frac{dx}{dv} = \frac{10v}{100 - v} = \frac{1000 - 10(100 - v)}{100 - v}$$

$$\text{So } \int dx = \int \frac{1000}{100 - v} - 10 dv$$

$$\text{So } x = -1000 \ln|100 - v| - 10v + c, \text{ for some constant } c$$

$$\text{But } x = 0 \text{ when } v = 0$$

$$\text{So } c = 1000 \ln 100 \text{ and from (iii) } v < 100.$$

$$\text{So } x = 1000 (\ln 100 - \ln(100 - v)) - 10v.$$

$$\text{So } x = 1000 \ln \left(\frac{100}{100 - v} \right) - 10v \text{ m, as required.} \quad \boxed{2}$$

(v) Let $v = 50$

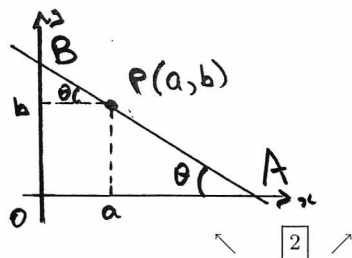
$$\text{So } x = 1000 \ln 2 - 500$$

$$\text{So } x = 500(\ln 4 - 1)$$

$$\text{So } x = 193.15$$

$$\text{Hence the object has fallen approximately } 193.15 \text{ metres.} \quad \boxed{1}$$

7. (a) (i)



$$\begin{aligned} AP &= b \operatorname{cosec} \theta \\ \text{and } PB &= a \sec \theta. \\ AB &= a \sec \theta + b \operatorname{cosec} \theta \end{aligned}$$

(ii) $\frac{d}{d\theta}(AB) = a \sec \theta \tan \theta - b \operatorname{cosec} \theta \cot \theta$

If $\frac{d}{d\theta} AB = 0$

then $a \sec \theta \tan \theta = b \operatorname{cosec} \theta \cot \theta$

So $\frac{\operatorname{cosec} \theta \cot \theta}{\sec \theta \tan \theta} = \frac{a}{b}$

So $\frac{\cot \theta}{\sec \theta \sin \theta \tan \theta} = \frac{a}{b}$

So $\frac{\cot \theta}{\tan^2 \theta} = \frac{a}{b}$

So $\cot^3 \theta = \frac{a}{b}$

Hence $\cot \theta = \left(\frac{a}{b}\right)^{\frac{1}{3}}$

Now $\frac{d^2}{d\theta^2} AB = a \sec \theta \tan^2 \theta + a \sec^3 \theta + b \operatorname{cosec} \theta \cot^2 \theta + b \operatorname{cosec}^3 \theta$

But $0 \leq \theta \leq \frac{\pi}{2}$ and hence all the trigonometric functions are positive so $\frac{d^2}{d\theta^2} AB > 0$.

So $\cot \theta = \left(\frac{a}{b}\right)^{\frac{1}{3}}$ minimises AB . 3

(iii) $\cot \theta = \frac{a^{\frac{1}{3}}}{b^{\frac{1}{3}}}$ and as θ is acute we can represent θ as shown in the right triangle.

Hence $r^2 = a^{\frac{2}{3}} + b^{\frac{2}{3}}$

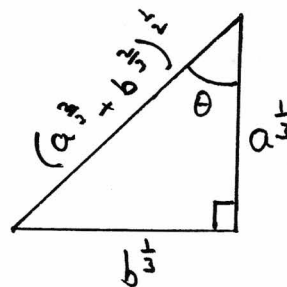
So $r = \left(a^{\frac{2}{3}} + b^{\frac{2}{3}}\right)^{\frac{1}{2}}$

Hence $\sec \theta = \frac{(a^{\frac{2}{3}} + b^{\frac{2}{3}})}{a^{\frac{1}{3}}}$ and $\operatorname{cosec} \theta = \frac{(a^{\frac{2}{3}} + b^{\frac{2}{3}})}{b^{\frac{1}{3}}}$

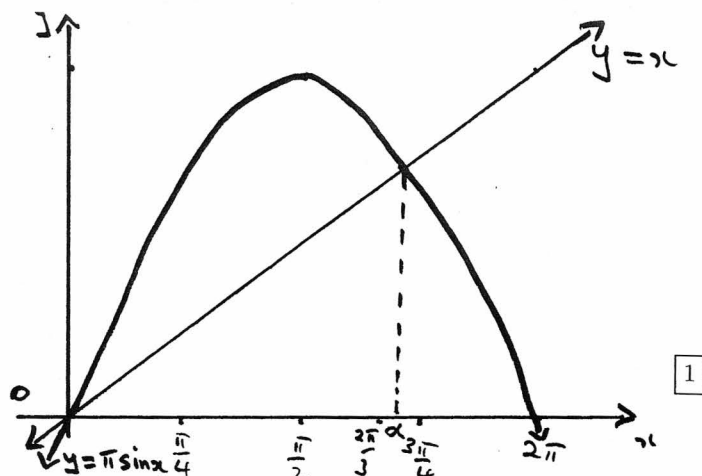
So the minimum length of AB is:

$$\begin{aligned} & a \frac{(a^{\frac{2}{3}} + b^{\frac{2}{3}})}{a^{\frac{1}{3}}} + b \frac{(a^{\frac{2}{3}} + b^{\frac{2}{3}})}{b^{\frac{1}{3}}} \\ &= \left(a^{\frac{2}{3}} + b^{\frac{2}{3}}\right)^{\frac{1}{2}} \left(a^{\frac{2}{3}} + b^{\frac{2}{3}}\right). \end{aligned}$$

Hence the minimum length of $AB = \left(a^{\frac{2}{3}} + b^{\frac{2}{3}}\right)^{\frac{3}{2}}$, as required. 2



(b) (i)



(ii) As $y = x$ and $y = \pi \sin x$ intersect there exists some value, α say such that $\pi \sin \alpha = \alpha$.

Consider the function $g(x) = \pi \sin x - x$.

$$\begin{aligned} g\left(\frac{2\pi}{3}\right) &= \pi \cdot \frac{\sqrt{3}}{2} - \frac{2\pi}{3} \\ &= \frac{3\sqrt{3} - 4}{6}\pi \approx 0.626 > 0. \end{aligned}$$

$$\begin{aligned} g\left(\frac{3\pi}{4}\right) &= \frac{\pi}{\sqrt{2}} - \frac{3\pi}{4} \\ &= \frac{1}{4}(2\sqrt{2} - 3)\pi \approx -0.135 < 0. \end{aligned}$$

So $g(x)$ being continuous between $\frac{2\pi}{3}$ and $\frac{3\pi}{4}$ and as $g\left(\frac{2\pi}{3}\right) \cdot g\left(\frac{3\pi}{4}\right) < 0$ there exists a zero α such that $\frac{2\pi}{3} < \alpha < \frac{3\pi}{4}$. 1

(iii) (α) $f(-x) = \sqrt{\pi^2 - (-x)^2} \cos(-x) - (-x) \sin(-x)$
 $= \sqrt{\pi^2 - x^2} \cos x - x \sin x$

Hence $f(-x) = f(x)$

That is $f(x)$ is even. 1

(β) $f(0) = \pi$.

$$f\left(\frac{\pi}{3}\right) = \sqrt{\pi^2 - \frac{\pi^2}{9}} \cdot \frac{1}{2} - \frac{\pi}{3} \cdot \frac{\sqrt{3}}{2}$$

$$f\left(\frac{\pi}{3}\right) = \frac{2\sqrt{2} - \sqrt{3}}{6}\pi \approx 0.574.$$

$$f\left(\frac{\pi}{2}\right) = \sqrt{\pi^2 - \frac{\pi^2}{4}} \cdot 0 - \frac{\pi}{2} \cdot 1$$

$$f\left(\frac{\pi}{2}\right) = -\frac{\pi}{2}.$$

$$f(\pi) = \sqrt{\pi^2 - \pi^2} \cdot -1 - \pi \sin \pi$$

$$f(\pi) = 0. \quad \text{[1]}$$

(γ) $f(\alpha) = \sqrt{\pi^2 - \alpha^2} \cos \alpha - \alpha \sin \alpha$

$$= \pi \sqrt{\cos^2 \alpha} \cos \alpha - \pi \sin^2 \alpha$$

But $\frac{2\pi}{3} < \alpha < \frac{3\pi}{4}$ so $\cos \alpha < 0$ hence $\sqrt{\cos^2 \alpha} = -\cos \alpha$

$$= \pi(-\cos^2 \alpha - \sin^2 \alpha)$$

$$\text{So } f(\alpha) = -\pi. \quad \text{[1]}$$

$$(\delta) f'(x) = \frac{1}{2} \frac{1}{\sqrt{\pi^2 - x^2}} - 2x \cos x - \sin x \sqrt{\pi^2 - x^2} - x \cos x - \sin x$$

$$\text{So } f'(x) = - \left(\frac{x \cos x}{\sqrt{\pi^2 - x^2}} + \sin x \sqrt{\pi^2 - x^2} + x \cos x + \sin x \right)$$

$$\text{So } f'(\alpha) = \frac{-\alpha \cos \alpha}{\sqrt{\pi^2 - \alpha^2}} - \frac{\sqrt{\pi^2 - \alpha^2} \sin \alpha}{1} - \alpha \cos \alpha - \sin \alpha$$

$$\text{That is } f'(\alpha) = \sin \alpha + \pi \cos \alpha \sin \alpha - \pi \cos \alpha \sin \alpha - \sin \alpha$$

$$\text{So } f'(\alpha) = 0.$$

Hence $x = \alpha$ is a stationary point.

$$\text{Now } \frac{\pi}{2} < \frac{2\pi}{3} < \alpha < \frac{3\pi}{4}.$$

$$\text{So } f'\left(\frac{\pi}{2}\right) = - \left(\frac{\sqrt{3}}{2} \pi + 1 \right) < 0$$

$$\text{and } f'\left(\frac{3\pi}{4}\right) = - \left(-\frac{3}{\sqrt{14}} + \frac{\sqrt{7}}{4\sqrt{2}} \pi - \frac{3}{4\sqrt{2}} \pi + \frac{1}{\sqrt{2}} \right) \approx 0.29 > 0$$

Hence $(\alpha, -\pi)$ is a minimum.

But $f(x)$ is even so $(-\alpha, -\pi)$ is a minimum.

As $f(x)$ is continuous there must be a maximum between the two minimums above. As $f(x)$ is even the only possible maximum must occur at $x = 0$. That is there is a maximum at $(0, \pi)$.

So the turning points and their nature are:

$$\begin{cases} (-\alpha, -\pi) & \text{minimum,} \\ (0, \pi) & \text{maximum,} \\ (\alpha, -\pi) & \text{minimum.} \end{cases} \quad \boxed{3}$$

[As a matter of interest $\alpha \approx 2.31373413208$.]

$$8. \text{ (a) (i) } \sin n\theta + \sin(n-2)\theta \\ = 2\sin(n-1)\theta \cos \theta$$

$$\text{Hence } k = n-1. \quad \boxed{1}$$

$$\begin{aligned} \text{(ii) } I_n + I_{n-2} &= \int (\sin n\theta + \sin(n-2)\theta) \sec \theta \, d\theta \\ &= 2 \int \sin(n-1)\theta \cos \theta \sec \theta \, d\theta \\ &= 2 \int \sin(n-1)\theta \, d\theta \\ &= -\frac{2}{n-1} \cos(n-1)\theta + C, \text{ for some constant } C. \end{aligned}$$

$$\text{So } I_n + I_{n-2} = \frac{2 \cos(n-1)\theta}{1-n} + C \text{ as required.} \quad \boxed{2}$$

$$\text{(iii) } \frac{\cos 5\theta \sin \theta}{\cos \theta} = \sec \theta \left(\frac{1}{2} \sin 6\theta - \frac{1}{2} \sin 4\theta \right).$$

$$\begin{aligned} \text{Now } \int_0^{\frac{\pi}{2}} \frac{\cos 5\theta \sin \theta}{\cos \theta} \, d\theta &= \frac{1}{2} \left[I_6 - I_4 \right]_0^{\frac{\pi}{2}} \\ &= \frac{1}{2} \left[I_6 + I_4 - 2I_4 \right]_0^{\frac{\pi}{2}} \\ &= \frac{1}{2} \left[I_6 + I_4 \right]_0^{\frac{\pi}{2}} - \left[I_4 \right]_0^{\frac{\pi}{2}} \\ &= \frac{1}{2} \left[\frac{\cos 5\theta}{-5} \right]_0^{\frac{\pi}{2}} - \left[I_4 + I_2 - I_2 \right]_0^{\frac{\pi}{2}} \\ &= \frac{1}{5} - \left[\frac{2 \cos 3\theta}{-3} \right]_0^{\frac{\pi}{2}} + \left[I_2 \right]_0^{\frac{\pi}{2}} \\ &= \frac{1}{5} - \frac{2}{3} + \left[I_2 + I_0 - I_0 \right]_0^{\frac{\pi}{2}} \\ &= \frac{1}{5} - \frac{2}{3} + \left[\frac{2 \cos \theta}{-1} \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} 0 \, d\theta \\ &= \frac{1}{5} - \frac{2}{3} + 2 - 0 \\ &= \frac{23}{15} \text{ as required.} \quad \boxed{4} \end{aligned}$$

(b) (i) $\psi(x) = a_1 + a_2 + \cdots + a_k + x - (k+1)(a_1 a_2 \cdots a_k x)^{\frac{1}{k+1}}.$

So $\psi'(x) = 1 - (a_1 a_2 \cdots a_k x)^{\frac{1}{k+1}-1} (a_1 a_2 \cdots a_k)$

$$\psi'(x) = 1 - (a_1 a_2 \cdots a_k)^{\frac{1}{k+1}} x^{\frac{1}{k+1}-1}$$

$$\psi'(x) = 1 - (a_1 a_2 \cdots a_k)^{\frac{1}{k+1}} x^{-\frac{k}{k+1}}$$

When $\psi'(x) = 0$ then

$$(a_1 a_2 \cdots a_k)^{\frac{1}{k+1}} x^{-\frac{k}{k+1}} = 1$$

$$\text{So } x^{-\frac{k}{k+1}} = (a_1 a_2 \cdots a_k)^{-\frac{1}{k+1}}$$

$$\text{So } x^k = (a_1 a_2 \cdots a_k)$$

Hence $\psi'(x) = 0$, when $x = (a_1 a_2 \cdots a_k)^{\frac{1}{k}} = x_0$.

$$\text{Now } \psi''(x) = \left(\frac{k}{k+1}\right) (a_1 \cdots a_k)^{\frac{1}{k+1}} x^{-\frac{2k+1}{k+1}}$$

$$\text{So } \psi''(x_0) = \frac{k}{k+1} (a_1 \cdots a_k)^{\frac{1}{k+1}} \left((a_1 \cdots a_k)^{\frac{1}{k}}\right)^{-\frac{2k+1}{k+1}}$$

$$\text{So } \psi''(x_0) = \frac{k}{k+1} (a_1 \cdots a_k)^{\frac{1}{k+1} - \frac{2k+1}{k(k+1)}}$$

$$\text{That is } \psi''(x_0) = \frac{k}{k+1} (a_1 \cdots a_k)^{-\frac{1}{k}} \text{ or } \frac{k}{(k+1)G_k} > 0, \text{ as } k, G_k > 0.$$

Hence the minimum value of $\psi(x)$ occurs at $x = x_0$. 3

(ii) Consider the proposition that

$$\text{"if } A_n = \frac{a_1 + a_2 + \cdots + a_n}{n} \text{ and } G_n = \sqrt[n]{a_1 a_2 \cdots a_n} \text{ then } A_n \geq G_n \text{"}$$

Now $A_1 = a_1$ and $G_1 = \sqrt[1]{a_1} = a_1$ hence $A_1 \geq G_1$.

Hence the proposition is true for $n = 1$.

Let k be some positive integer such that the proposition is true.

That is $A_k \geq G_k$.

From (i) $\psi(a_{k+1}) \geq \psi(x_0)$.

$$\text{That is } a_1 + a_2 + \cdots + a_k + a_{k+1} - (k+1)(a_1 a_2 \cdots a_{k+1})^{\frac{1}{k+1}}$$

$$\geq a_1 + a_2 + \cdots + a_k + G_k - (k+1)(a_1 a_2 \cdots a_k G_k)^{\frac{1}{k+1}}.$$

$$\left((a_1 a_2 \cdots a_k G_k)^{\frac{1}{k+1}}\right) = \left((a_1 \cdots a_k)^{1+\frac{1}{k}}\right)^{\frac{1}{k+1}} = \left((a_1 \cdots a_k)^{\frac{k+1}{k}}\right)^{\frac{1}{k+1}} = G_k$$

$$\text{That is } (k+1)(A_{k+1} - G_{k+1}) \geq kA_k + G_k - (k+1)G_k$$

$$\text{So } (k+1)(A_{k+1} - G_{k+1}) \geq k(A_k - G_k) \geq 0$$

Hence $A_{k+1} \geq G_{k+1}$.

As $A_1 \geq G_1$ and $A_k \geq G_k$ implies $A_{k+1} \geq G_{k+1}$ for some positive integer k then by the principle of mathematical induction $A_n \geq G_n$ for all positive

integers n . 5